Mingyan Wu

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Research Interests: LLM Reasoning, Retrieval-Augmented Generation (RAG), LLM-based Agents

Education

M. Sc.: ***Computer Science*** **09.2024 - 06.2027**
Northeastern University *Shenyang, China*

B. Sc.: ***Computer Science*** **09.2020 - 06.2024**
Yangtze University | GPA Ranking: 2.7% *Jingzhou, China*

Publication

* Equal contribution.

- **Mingyan Wu**, Zhenghao Liu, Yukun Yan, Xinze Li, Shi Yu, Zheni Zeng, Yu Gu, and Ge Yu. 2025. **RankCoT: Refining Knowledge for Retrieval-Augmented Generation through Ranking Chain-of-Thoughts. ACL 2025 Main Conference.** Paper Link. Code Link.
- **Mingyan Wu***, Zhenghao Liu*, Xinze Li, Yuqing Lan, Yukun Yan, Shuo Wang, Cheng Yang, Minghe Yu, Zheni Zeng, and Maosong Sun. 2026. **Finding What Matters: Anchoring Context Knowledge with Evolving Indices for Iterative Retrieval. To appear in EMNLP 2026.** Paper Link. Code Link.
- **Mingyan Wu***, Han Yang*, Omer Ben-Porat, Yftah Ziser. 2026. **When Knowledge Is Not Free: Cost-Aware Evidence Selection in Retrieval-Augmented Generation. To appear in Findings of EMNLP 2026.** Paper Link. Code Link.
- Han Yang*, **Mingyan Wu***, Bailan He, Zeyu Cao, Sikuan Yan, Kevin Qinghong Lin, Zifeng Ding. 2026. **Reasoning Compression with Mixed-Policy Distillation.** arXiv:2605.08776. Paper Link.
- Boer Zhang, **Mingyan Wu**, Dongzhuoran Zhou, Yuqicheng Zhu, Wendong Fan, Puzhen Zhang, Zifeng Ding, Guohao Li, and Yuan He. 2026. **EigentSearch-Q+: Enhancing Deep Research Agents with Structured Reasoning Tools. ACM CAIS 2026 Demo.** Paper Link.

Internship Experiences

THUNLP Lab, Tsinghua University | Supervised by Dr. Yukun Yan **04/2024 – 08/2026**
Research Intern *Beijing, China*

- Project 1: Knowledge Refinement for RAG | Related Paper: RankCoT (ACL 2025)
  - Investigated how RAG systems can reliably identify and use relevant evidence from noisy retrieval results during multi-step reasoning
  - Proposed **RankCoT**, a ranking-based **knowledge refinement** framework, as a solution to improve reliability against noisy documents
  - Conducted **post-training of LLMs** with the **DPO algorithm** to improve the ability to identify and refine knowledge
  - Designed and conducted multiple experiments across multiple knowledge-intensive QA benchmarks
- Project 2: Maintaining Focus across Steps in Long-term Reasoning | Related Paper: KAIR (EMNLP 2026)
  - Investigated distraction of RAG systems across steps in long-term reasoning tasks
  - Proposed **KAIR**, a **K**nowledge **A**nchoring framework for **I**terative **R**etrieval that **anchors knowledge** within retrieved knowledge to guide LLMs to **locate the key information** during iterative retrieval and answer generation.
  - Conducted experiments across multiple benchmarks
- Other Contributions: Drafting preliminary documents, guiding junior students

Researches & Projects

Cost-Aware Evidence Selection for Retrieval-Augmented Generation **02/2026 – 08/2026**
Research Collaboration with Prof. Dr. Yftah Ziser, NVIDIA Research, University of Groningen

- Studied evidence selection for retrieval-augmented generation under **explicit information-acquisition costs**.
- Developed and implemented a **cost-aware selection framework** that balances the utility of additional evidence against its acquisition cost.
- Related publication: **Cost-Aware RAG (EMNLP 2026 Findings)**.

Reasoning Compression for LLMs **01/2026 – present**

Research Collaboration with Dr. Zifeng Ding, University of Cambridge, Mina AI

- Contributed to a **Mixed-Policy Distillation (MPD)** framework that transfers reasoning behavior from multiple teacher-generated policies into a more efficient student model.
- Participated in method implementation and experimental evaluation across reasoning benchmarks.
- Related publication: **Reasoning Compression with MPD (arXiv:2605.08776)**.

Structured Reasoning for Deep-Research Agents | CAMEL-AI **08/2025 – 05/2026**

- Co-developed **EigentSearch-Q+**, which augments **Eigent's browser agent** with structured tools for query planning, research-progress monitoring, and evidence extraction from long web pages.
- Evaluated Q+ across four deep-research benchmarks and multiple LLM backends, demonstrating improved search trajectories and accuracy over the original Eigent browser agent.
- Developed the **evaluation framework** for EigentSearch-Q+, enabling reproducible **benchmarking** across six datasets: SimpleQA, SimpleQA-Verified, BrowseComp, WebWalkerQA, MuSiQue, and FRAMES.
- Project Link: https://github.com/camel-ai/eigent_search
- Related publication: **EigentSearch-Q+ (ACM CAIS 2026 Demo)**.

UltraRAG 2.0 Framework Development | OpenBMB **05/2025 – 09/2025**

- Contributed to UltraRAG 2.0, a modular **MCP-based** framework for building and evaluating **complex RAG pipelines**.
- Implemented and integrated reproducible pipelines for **RankCoT, IRCoT, and Search-o1**, adapting their multi-stage retrieval and reasoning workflows to UltraRAG's MCP-based architecture.
- Project Link: https://github.com/OpenBMB/UltraRAG **Stars: 5.7k**

Honors & Awards

First-Class Graduate Scholarship, Northeastern University — 2024, 2026

Academic Service

- Reviewer, ACL Rolling Review (ARR) — 2025–2026
- Reviewer, Conference on Language Modeling (COLM) — 2026

Technical Skills

- Programming: Python, Bash, C, MATLAB
- NLP: LLM Fine-Tuning, LLM Inference, Corpus Indexing, Information Retrieval, LLM Evaluation
- Infrastructure: vLLM, DeepSpeed, Hugging Face TRL
- Tools: Git, Linux, LaTeX, Markdown